m17web User Manual

M17 digital voice on a Raspberry Pi with an SDR, driven from your web browser

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This manual is for anyone using the radio. It explains what the program does, how to start it, how to make a contact, and what to do when something does not work. You do not need to understand M17, SDR or Linux to use it — if someone set the system up for you, start at section 4.

A separate README in the same folder covers the technical details: building the program, the audio back ends, and the transmit timing.

What is in this manual

	Setting it up		Using it
1	What m17web does	7	Audio: where your voice comes from and goes
2	What you need	8	The controls, panel by panel
3	Starting and stopping it	9	Commands
4	Opening the interface	10	If something goes wrong
5	The screen at a glance	11	Good practice
6	Making a contact	12	Quick reference

1. What m17web does

M17 is a digital voice mode for amateur radio: like FM voice, but it sends your speech as data, which gives clear audio and carries your callsign with it. m17web is a program that runs on a small computer (a Raspberry Pi) connected to a software-defined radio. It sends and receives M17 voice, and you control it entirely from a web page — no terminal commands needed in normal use.

Your voice comes from a microphone plugged into the Pi, or from the computer you are browsing from. Received voice comes out of the Pi's speaker (or your computer's).

If you are only practising, the program can run without a radio, in a mode that feeds your own audio straight back. See section 9.

2. What you need

Item	Notes
Raspberry Pi 4	Any 4 GB or 8 GB model, with a network connection. The Pi runs m17web.
ADALM-Pluto (PlutoSDR / Pluto+)	The radio itself, connected to the Pi by USB. A LimeSDR also works.
USB sound card with microphone and speaker or headphones	For audio at the Pi. Recommended - it is the most reliable audio path.
Antenna and a valid amateur licence	You need a licence for any frequency you transmit on.
A web browser	On any computer, tablet or phone on the same network as the Pi.

3. Starting and stopping it

m17web is started from a terminal on the Pi. If someone else set the system up for you, it may already be running — in that case go straight to section 4.

Start it (this also works after a reboot):

```
cd /mnt/hf/shared/projects/m17_sdr_web/m17gismo
RADIO=pluto MODE=radio bash tools/pi_m17web.sh start -a alsa
```

It keeps running after you close the terminal window. **Check it:**

```
bash tools/pi_m17web.sh status
```

Stop it, or look at the log to see what it is doing:

```
bash tools/pi_m17web.sh stop
bash tools/pi_m17web.sh log
```

What the settings mean

Setting	What it does
RADIO=pluto	Use the PlutoSDR. Use **RADIO=sim** instead to practise with no radio attached.

Setting	What it does
MODE=radio	Radio in, local sound out — the normal voice mode.
-a alsa	Send the voice to the Pi's sound card (see section 7). Use **-a browser** to use the browser's own microphone and speakers instead.

4. Opening the interface

In a browser on the same network, go to the address of the Pi followed by the port number:

```
http://10.168.1.33:8080/
```

Substitute the address of your own Pi. Bookmark it: this is the whole interface.

At the top right there is an **online / offline** marker. If it says offline, the page has lost contact with the Pi and nothing you press will do anything — check that the program is running (section 3) and reload the page.

5. The screen at a glance

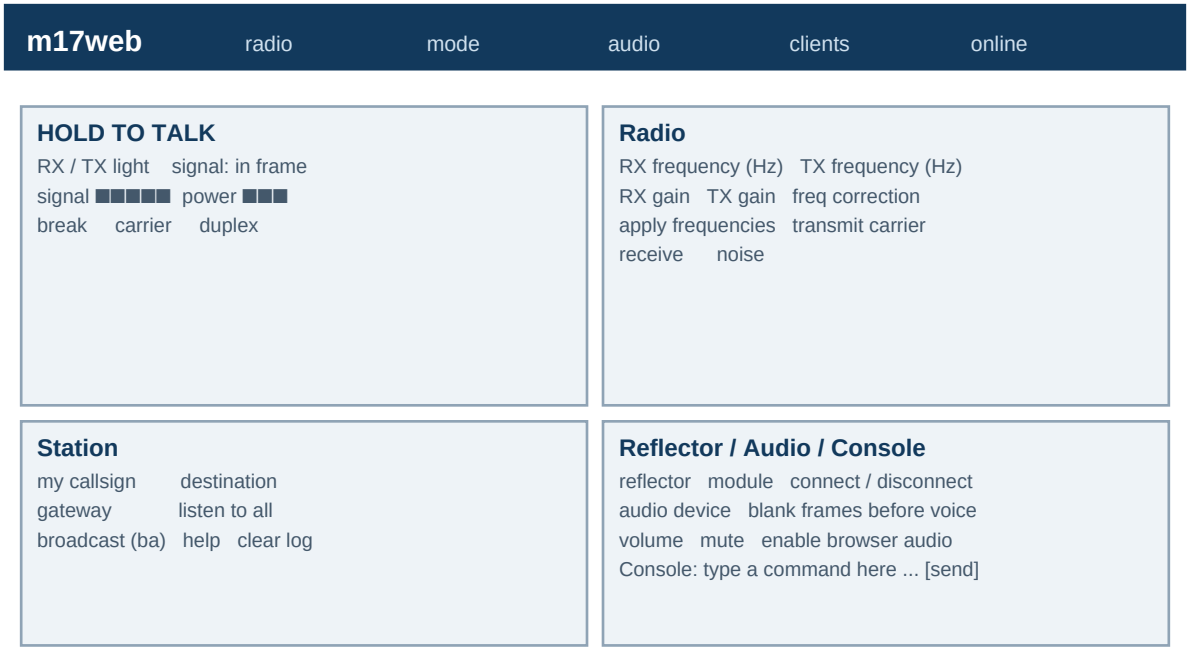

Figure 1: the panels on the page. The header is always visible; PTT, Station and Radio on the left, Radio, Reflector, Audio and Console on the right. (Sketch, not a screenshot.)

Area	What it is for
HOLD TO TALK	Press and hold while you speak, then release. You can hold the space bar instead.
RX / TX light	Shows whether you are transmitting. Next to it, **in frame** means a valid M17 signal is being decoded, **no signal** means nothing is there.
signal and power bars	How strong the received signal is, and your transmit power setting.
Station	Your callsign, who you are calling, and the broadcast and log buttons.
Radio	The frequencies and gain settings, and the button that applies them.
Reflector	Linking over the internet to other M17 users (you may never need it).
Audio	Where your voice comes from and goes, the volume, and how healthy your transmit audio is.
Console	A command box for advanced settings, and the log that shows what is happening.

6. Making a contact

1. Type your callsign into **Station** → **my callsign**. It is sent with every transmission, so it must be correct.
2. Agree a frequency with the other station, then type it into **Radio** → **RX frequency (Hz)** and press **apply frequencies**. This sets transmit to the same frequency.
3. Listen first. When the other station speaks, the light shows **RX**, the signal bar moves, **in frame** appears and you hear them.

4 To reply, hold **HOLD TO TALK**, speak in your normal voice a little slower than a telephone call, then release.

5 Wait for the other station to finish before you press again — while you are transmitting, you cannot hear them.

Only transmit on frequencies your licence allows, and check what is already in use before you press. If you are testing, a dummy load or the lowest gain keeps you out of trouble.

7. Audio — where your voice comes from and goes

Set this once in the **Audio** panel; it stays set.

Option	What it means	When to use it
Raspberry Pi sound card (direct)	Uses the microphone and speaker plugged into the Pi. Talks to the sound card directly.	**Recommended.** The most reliable choice.
browser microphone / speaker	Uses the microphone and speakers of the computer you are browsing from, sent over the network.	When you are away from the Pi, or the Pi has no sound card. See the note below.
Raspberry Pi sound card (PulseAudio)	The same card, but through the PulseAudio sound service.	Only if something else on the Pi needs to share the card. Direct is more reliable.
browser + Pi sound card	Both at once, listening in two places.	Rarely needed.
mute (no audio)	Silence going out and coming in.	For testing.

A change takes effect at the start of your next transmission; the panel shows **(applying...)** until then.

If the browser refuses your microphone

Browsers only allow microphone access on a secure page. A page opened as **http://<address>:8080/** is not secure, so the microphone may be blocked. Either open **http://localhost:8080/** in a browser on the Pi itself, or use the Pi's sound card option instead — which avoids the problem completely and is what most users want.

Blank frames before voice, and the transmit health line

Next to the audio device there are two things worth knowing about.

blank frames before voice is the amount of silence sent at the start of each transmission, before your speech. It gives the receiving station time to lock onto your signal, so your first word is not clipped. The default is 4 (about 80 thousandths of a second). If people tell you the start of your first word is missing, set it to 6 or 8.

The line beside it — **transmit clean**, or **dropped N samples** — is the health of your transmit audio. **Transmit clean** means the Pi kept up with the radio for the whole transmission. If it reports dropped samples, raise the blank frames a little and try again; if it keeps reporting

drops, the Pi is too busy (close other programs) or something needs looking at by whoever maintains the system.

volume and **mute** affect only what you hear in the browser. The small **mic** and **spk** bars show the audio levels going out and coming in — the mic bar should move when you speak.

8. The controls, panel by panel

Station

Control	What it does
my callsign	Your callsign. Sent with every transmission.
destination	Who you are calling. Leave as ALL unless you are calling one specific station.
gateway	Used when linking through a reflector or repeater (advanced).
listen to all	Receive everything on the frequency, whoever it is addressed to.
broadcast (ba)	Address the transmission to everyone (the usual choice on a calling channel).
help / clear log	Lists every command in the log, or empties the log.

Radio

Control	What it does
RX frequency (Hz)	The frequency you listen on, in hertz (145500000 = 145.5 MHz).
TX frequency (Hz)	The frequency you transmit on. Usually the same as RX.
RX gain / TX gain	Receive and transmit levels. Leave them as they are unless someone asks you to change them.
freq correction	Fine tuning, for when a station sounds slightly off. Change it in small steps.
apply frequencies	Press this after changing a frequency or a gain, or nothing changes.
transmit carrier	Sends a plain carrier with no voice - for tests and alignment only.
receive	Returns to receiving if a transmission is stuck on, or after a test.
noise	Practice mode only: adds noise to the simulated signal.

Reflector

Reflectors link M17 stations over the internet, so you can talk to people out of radio range. Choose a reflector from the list, choose the channel letter (the **module**), and press **connect**. Press **disconnect** when you have finished — it is good manners, and it frees the channel for others.

Reflector use may need the gateway address filled in as well, and a working internet connection on the Pi. If you only want local radio contacts, ignore this panel.

Audio

Control	What it does
audio device	Where your voice comes from and goes - see section 7.
blank frames before voice	Silence sent before your speech at the start of a transmission. Default 4.
enable browser audio	Must be pressed once (and permission given) before the browser can play or record audio.
volume / mute	What you hear in the browser.

Control	What it does
mic / spk bars	Audio levels going out and coming in.

Console

Every setting also has a typed command, and the console is where you type them. You do not need it for normal operation; the help button lists the commands, and section 9 lists them all. The log below it shows what the program is doing — useful when something does not work.

9. Commands

Typed into the Console box, or into the program's own terminal. Type one command per line.

Command	What it does
rf 145500000	Set the receive frequency, in hertz.
tf 145500000	Set the transmit frequency, in hertz.
rg 0.5 / tg 0.5	Set the receive or transmit gain, from 0.0 to 1.0.
fc 1.0001	Frequency correction, from 0.9 to 1.1.
afc on / off	Automatic frequency control.
tx / rx	Start or stop transmitting (the PTT button does the same).
td / tc	Transmit duplex, or transmit a plain carrier.
sa G4ABC	Source address: your callsign.
da ALL	Destination address: who you are calling.
ga G4ABC G	Gateway address.
ba	Broadcast: call everyone.
la on / off	Listen to all addresses, or only yours.
conn NAME C	Connect to a reflector, on module C.
disc	Disconnect from the reflector.
mode radio	Normal mode: radio in, sound out. Alternatives: **loop** (audio loops back for practice), **gate** (radio to reflector), **client** (sound to reflector).
audio alsa	Voice through the Pi sound card. Alternatives: **browser**, **pulse**, **both**, **none**.
lead 4	Blank codec frames sent before the voice - see section 7.
load FILE	Run a list of commands from a file.
w	Clear the log pane.
b	Break: stop transmitting now.
q	Quit the program.
h	List the commands.

10. If something goes wrong

What you see	What to try
The page will not open at all	Check the address and the port (8080), and that the Pi is switched on and on the network. On the Pi, run **bash tools/pi_m17web.sh status**.
It says **offline** at the top	The page lost contact with the Pi. Reload the page. If it stays offline, check the program is still running on the Pi.
Nothing comes out of the speaker	In Audio, make sure the Pi's sound card is selected, the volume is up and mute is off, and that the speaker is plugged into the sound card (not the Pi's headphone socket).
The other station says my audio breaks up, or starts late	Look at the transmit health line in the Audio panel. Raise **blank frames before voice** to 6-8 and try again. If it still reports dropped samples, the Pi is struggling with something else running.
I hear nothing when the other station speaks	Check you are on the same frequency, and that **in frame** appears when they talk. If the signal bar moves but there is no audio, try **afc on**, then nudge **freq correction** slightly.
The browser will not use my microphone	Browsers only allow the microphone on a secure page. Use **http://localhost:8080/** on the Pi itself, or select the Pi's sound card instead (section 7).
The light never goes to TX	If the header says offline, the page cannot reach the Pi. Otherwise something else may be using the radio — stop and start the program.
The sound card is reported as busy	Another program (or an earlier run) is using it. Stop other audio programs, then stop and restart m17web.
The program will not start (port already in use)	An older copy is still running: **bash tools/pi_m17web.sh stop**, wait a moment, then start it again.
The radio cannot be found at all	The Pluto re-appears as a USB network interface when it is plugged in, and it needs the address 192.168.2.2. Check it is connected, then ask whoever maintains the system — this is covered in the README.

11. Good practice

- Transmit only on frequencies your licence covers, and at the lowest power that works.
- Listen before you transmit, and give way to a station already in contact.
- Keep your callsign correct in the Station panel — it is how other operators know who is calling.
- When using a reflector, say where you are going and disconnect when you have finished.
- For tests, use a dummy load or a carrier into a low gain rather than transmitting into the antenna.
- If you leave the system running, mute the audio if you do not want your room heard on the air.

12. Quick reference

Start it

```
cd /mnt/hf/shared/projects/m17_sdr_web/m17gismo
RADIO=pluto MODE=radio bash tools/pi_m17web.sh start -a alsa
```

Open it

```
http://10.168.1.33:8080/ (your own Pi's address)
```

Make a contact

Step	Do this
1	Station → my callsign: type your callsign.
2	Radio → RX frequency (Hz): type the agreed frequency, press **apply frequencies**.
3	Listen until you hear the other station (**in frame** appears).
4	Hold **HOLD TO TALK**, speak, release.
5	Let the other station finish before pressing again.

The three things that put most people right

If	Then
You cannot hear anything	Check the frequency, and whether **in frame** appears when the other station speaks. Try **afc on**.
Nobody can hear you	Look at the transmit line in the Audio panel. If it reports dropped samples, raise **blank frames before voice** to 6-8.
The page will not reach the Pi	On the Pi: **bash tools/pi_m17web.sh status**, then reload the page.

Settings worth knowing: **audio device** (use the Pi's sound card, direct), **blank frames before voice** (default 4) and **volume** in the Audio panel; **help** in the Station panel lists every command.

13. Where this came from, and further reading

m17web is a modification of **G4GUO/m17_sdr**, with the original terminal interface replaced by the web page described here. The M17 protocol it speaks is an open standard from the M17 Project; the voice coding is Codec2 by David Rowe and the M17 Project's Codec2-mod build. It uses the open-source libiio, ALSA and Codec2 libraries.

For the technical details — building it yourself, the audio back ends, the transmit timing and the rate planning — see **README.md** next to this manual, in the same folder.

The upstream project has no licence file, so this fork adds its own documentation only. Treat the software as you find it: it works, but check it against your own licence conditions before putting it on the air.

Notes

Frequencies in use locally, reflector names, who to call when it breaks: